

Daniel Hardesty Lewis

ML Engineer – distributed training infrastructure, cloud-scale ML systems, tabular data

Specializing in scalable cloud infrastructure for ML training and serving. Delivered production systems across tabular, geospatial, and time-series workloads at supercomputer scale.

Core Expertise

- 8 years **ML Engineering**: Design and operate ML training and inference infrastructure at scale, from data ingestion through model deployment, monitoring, and retraining
- 8 years **Backend & Data**: Build schemas, ETL, and production APIs (Python/FastAPI); manage data-intensive systems with Postgres, Redis, and Elasticsearch
- 8 years **Distributed Systems**: Scale ML workloads across distributed cloud and HPC systems using containerization and orchestration (Docker, Kubernetes, Ray), with explicit cost-performance tradeoffs
- 10 years **Geospatial AI**: Spatiotemporal modeling for risk, planning, and hazard decision-support use cases
- 8 years **Technical Leadership**: Collaborate across engineering, research, and product to ship; mentor engineers through design reviews, code reviews, and instruction

Technologies

Modeling	PyTorch; TensorFlow; scikit-learn; Transformers; LoRA/QLoRA/PEFT
LLM apps	embeddings; RAG; LangChain; LangGraph; DSPy; FAISS
Data stack	Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; BigQuery; Redis; Elasticsearch
Infrastructure	Linux; Git; Docker; Kubernetes; Terraform; Triton; Ray; Dask; Airflow; CI/CD; monitoring & observability
Additional	CUDA; C++; TypeScript; React

Experience

Professional

- Apr. 2025 – Present **Co-founder**, [Homecastr](#), New York, NY
 - Consumer real-estate pricing and forecasting platform with interactive uncertainty fancharts and temporal leakage-safe expanding-window evaluation; owned end-to-end workflow from source data to web delivery.
 - Architected a nationwide neighborhood-level diffusion forecasting model, **matching Zillow's 7.3% 0-year error benchmark at a 1-year forecast horizon** and maintaining a **35.6% median error over 4 years**.
 - Engineered architecture incorporating **spatial inducing tokens**, **per-horizon loss normalization**, and **DDIM sampling** to balance predictive precision with calibrated price trajectories.
 - Developed predictive validation optimizing PIT uniformity, interval coverage, and sharpness to **rigorously validate forecast reliability** before production delivery.
 - Owned the end-to-end pipeline from data ingestion to real-time serving and refined product strategy through direct feedback from **a16z** and **MetaProp**.
- Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, [Texas Advanced Computing Center \(TACC\)](#), Austin, TX
 - Competed in [DARPA World Modelers](#) to improve disaster resiliency and food security modeling in East Africa.
 - Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
 - Received research recognition from AAAI President Yolanda Gil during her [farewell address](#) for contributions to automated scientific modeling.

Research

- May 2025 – Aug. 2025 **Research assistant to Dir. Electrical Engineering Zoran Kostic**, [Columbia University](#), New York, NY
 - Math benchmarking and finetuning of Deepseek's GRPO algorithm
- Jan. 2025 – Oct. 2025 **Research assistant to Dir. Financial Engineering Ali Hirs** (**Industry-sponsored research**), [Columbia University](#), New York, NY
 - Developed a multi-asset CVAE latent factor model for **tabular financial data** with Skew-T Mixture priors; achieved 85% R^2 vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
 - Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.

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July 2019 – **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA

- Integrated hydrological models into the [MINT Platform](#) during a year-long competition with MITRE to provide DARPA the "Airflow for science"

Teaching

Jan. 2018 – **Co-instructor**, *The University of Texas at Austin*, Austin, TX

- Dec. 2020 ○ Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, and HPC.

Select Publications

- 2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TiiS*, 2021. DOI: [10.1145/3453172](#)
- 2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: [10.1145/3308557.3308711](#)

Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

Key Projects

- oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

Activities

Sailing Volunteer Atlantic SAR of the [Cheeki Rafiki](#)

Bikepacking US West & East Coasts

Languages

Spanish Native Proficiency

French Professional Proficiency

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